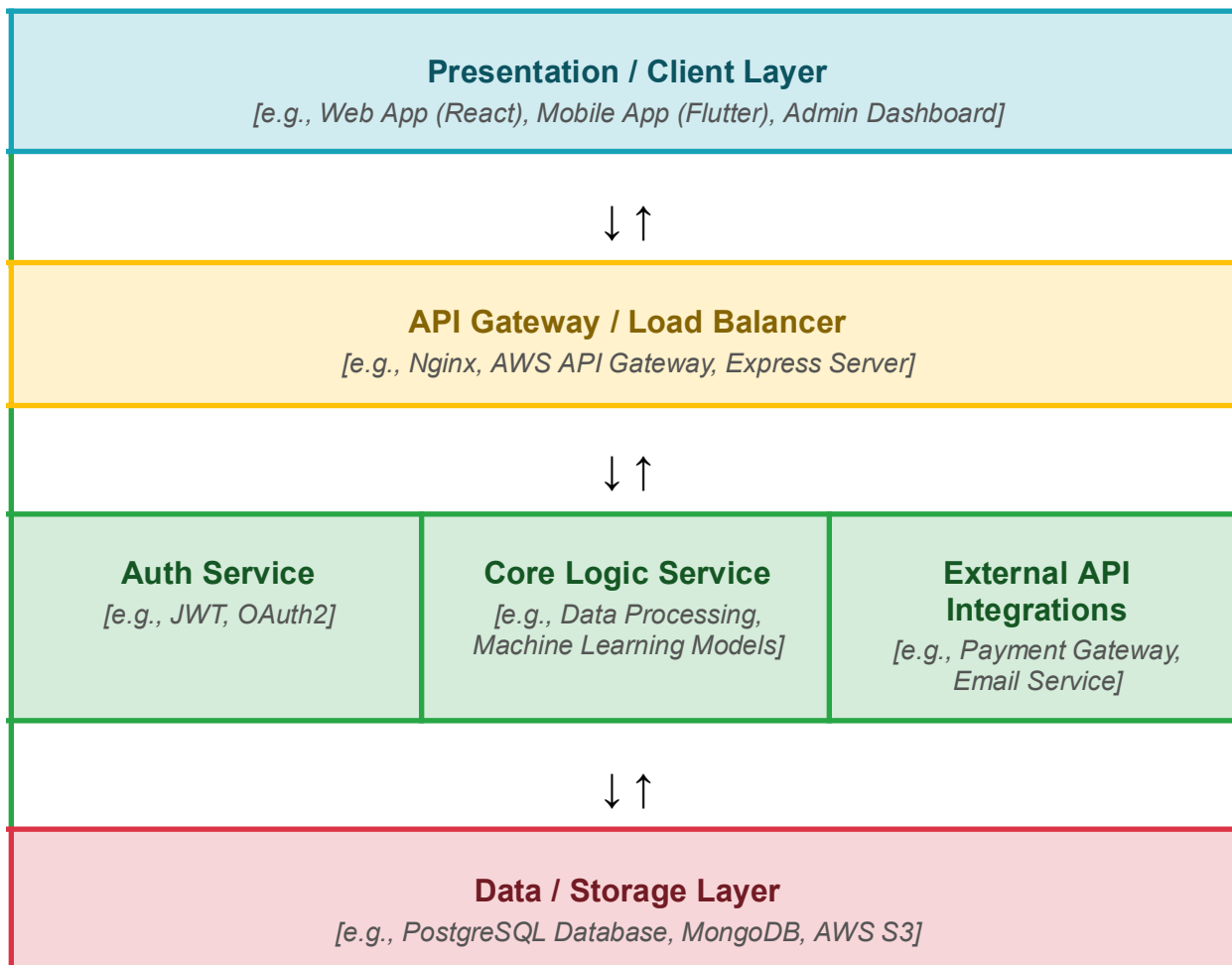


Solution Architecture

Date	02 July 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	5 Marks

Solution Architecture Diagram:

Below is the high-level visual representation of the solution's architecture, mapped out across the Presentation, Application, and Data layers.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Acts as the user-facing interface. It captures user inputs (Life Expectancy, Schooling Years, GNI), displays the predicted HDI tier, and renders interactive data visualizations and dashboards.	HTML5, CSS3, Bootstrap, JavaScript, Jinja2
API Gateway	Serves as the entry point for all client requests. It routes HTTP requests to the appropriate backend logic, handles URL routing, manages sessions, and returns formatted responses to the frontend.	Flask Framework, Gunicorn
Microservices	Contains the core business logic and processing engines. Includes the ML Engine (for data preprocessing and Random Forest inference), Auth Module (for secure login), and Visualization Module (for generating Matplotlib/Seaborn charts).	Python, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn
Database	Persists structured data required for the application's operation. Stores user authentication details, logs of past predictions, and references to the historical HDI dataset used for model training.	SQLite, SQLAlchemy (ORM)